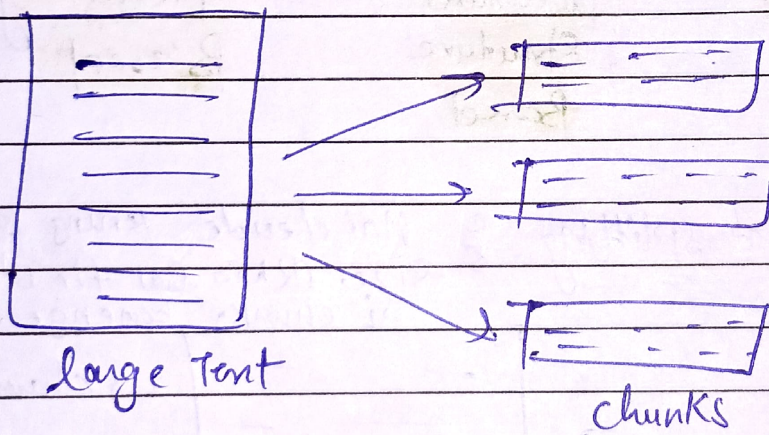


- ⑥ CSV loader → is a document loader used to load CSV files into langchain Documents objects - one per row, by default.

same code on GitHub - - - - -

- Tent Splitters $\frac{9}{6}$ Text splitting is the process of breaking large chunks of text (like Articles, PDFs, HTML Pages or books) into smaller, manageable pieces (chunks) that an LM can handle effectively.

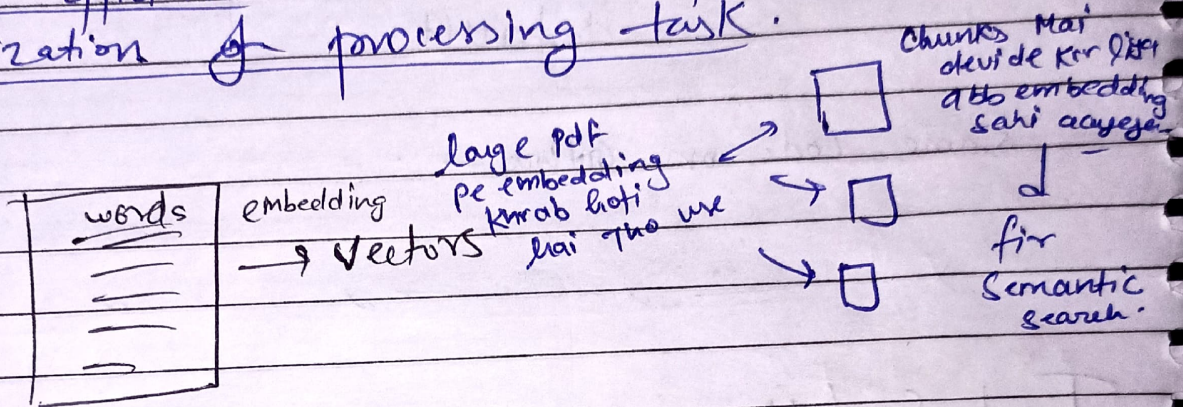


LM → content length
↓
50K Token → words
↑
Pplf → summarize
↓
1 lakh word.
no summarize.

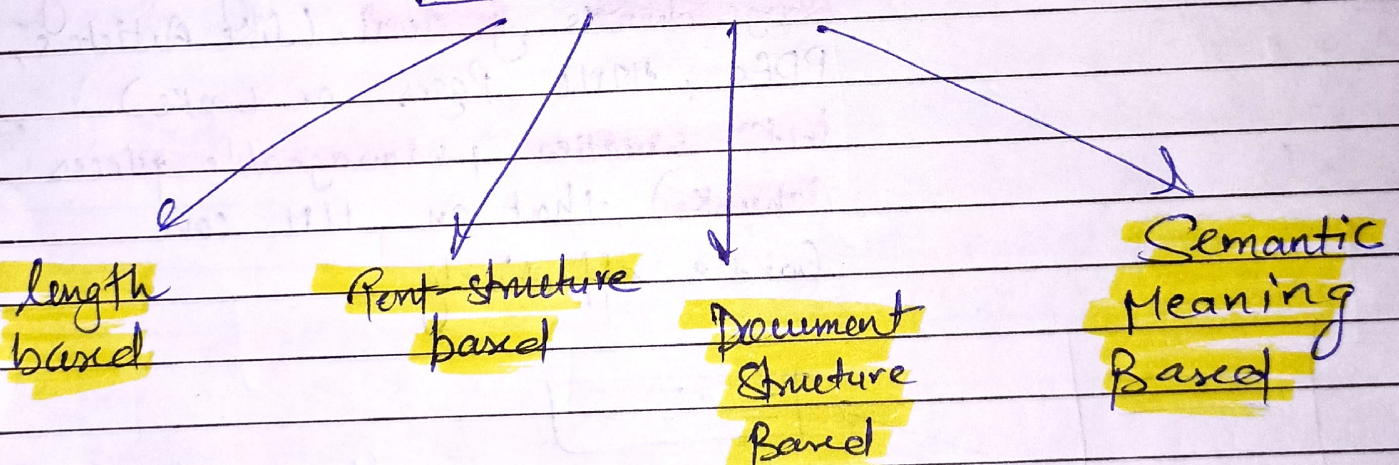
- Overcoming Model Limitations :- Many embeddings model and language models have maximum input size constraints, splitting allows us to process documents that would otherwise exceed these limits.

- Downstream tasks → embeddings, Semantic Search, Summarization

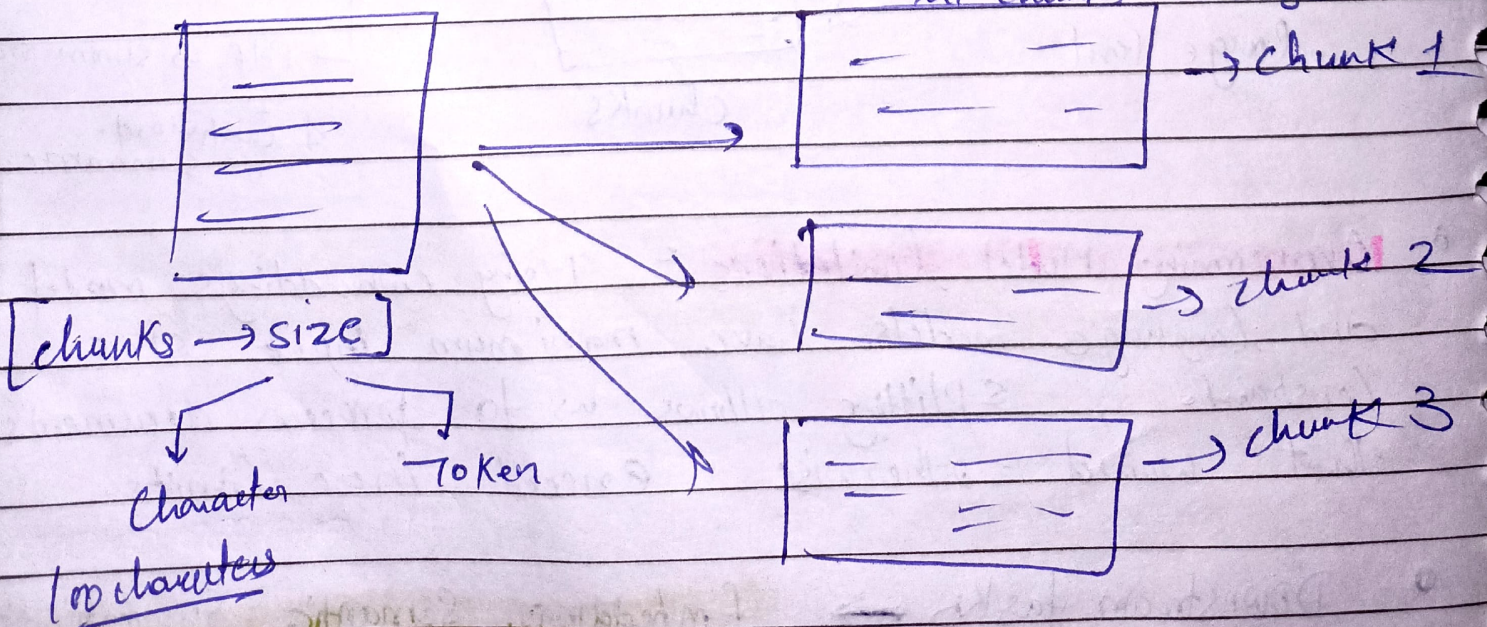
- o optimizing computational resources → working with smaller chunks of text can be more memory-efficient and allow for better parallelization of processing task.



Text splitters



① Length based Text splitting 4 8 Mai decide krung kitna size rkha hai fir use hi chunks banenge.



Disadvantage :- No semantic meaning.
 Kahi bhi Ruk Jayega Kyunki 100 words
 Set hai tho 100 words complete hote
 hi Kahi bhi ruk Jayega !!

Code :-

```
from langchain_text_splitters import CharacterTextSplitter
from langchain_community.document_loaders import PyPDFLoader
```

```
loader = PyPDFLoader("file path")
```

```
docs = loader.load()
```

```
splitter = CharacterTextSplitter (
```

```
    chunk_size = 200,
```

```
    chunk_overlap = 0, →
```

```
    separator = " "
```

(Bataiye ki character ke
 bich overlap kab hoga
 chunks mai !!)

```
)
```

```
result = splitter.split_documents(docs)
```

↳ document ko split krna !!

```
print(result[0].page_content)
```

[Note]

→ Jab hum chunk_size = 100 ya jitna Margi rkhate hai
 tho wo paragraph hai ya Pdf mai 100 words pe hi
 aake ruk Jayega wo ye nhi dekhga ki Koi
 meaning bar rha hai ki nhi wo bss 100 words pe aake
 ruk Jayega chunk_overlap ki help se hum ye problem
 solve kr skte hai !!

Chunk size Jitna reduce Krunga
Utna Chunks bante Jayenge

Date

⑤ Tent-Structured Based

[My name is shubham
I live in Delhi
How are you]

→ Structure
Sentences
words

→ Recurse
Tent
Splitter
Technique
Hehete hai.

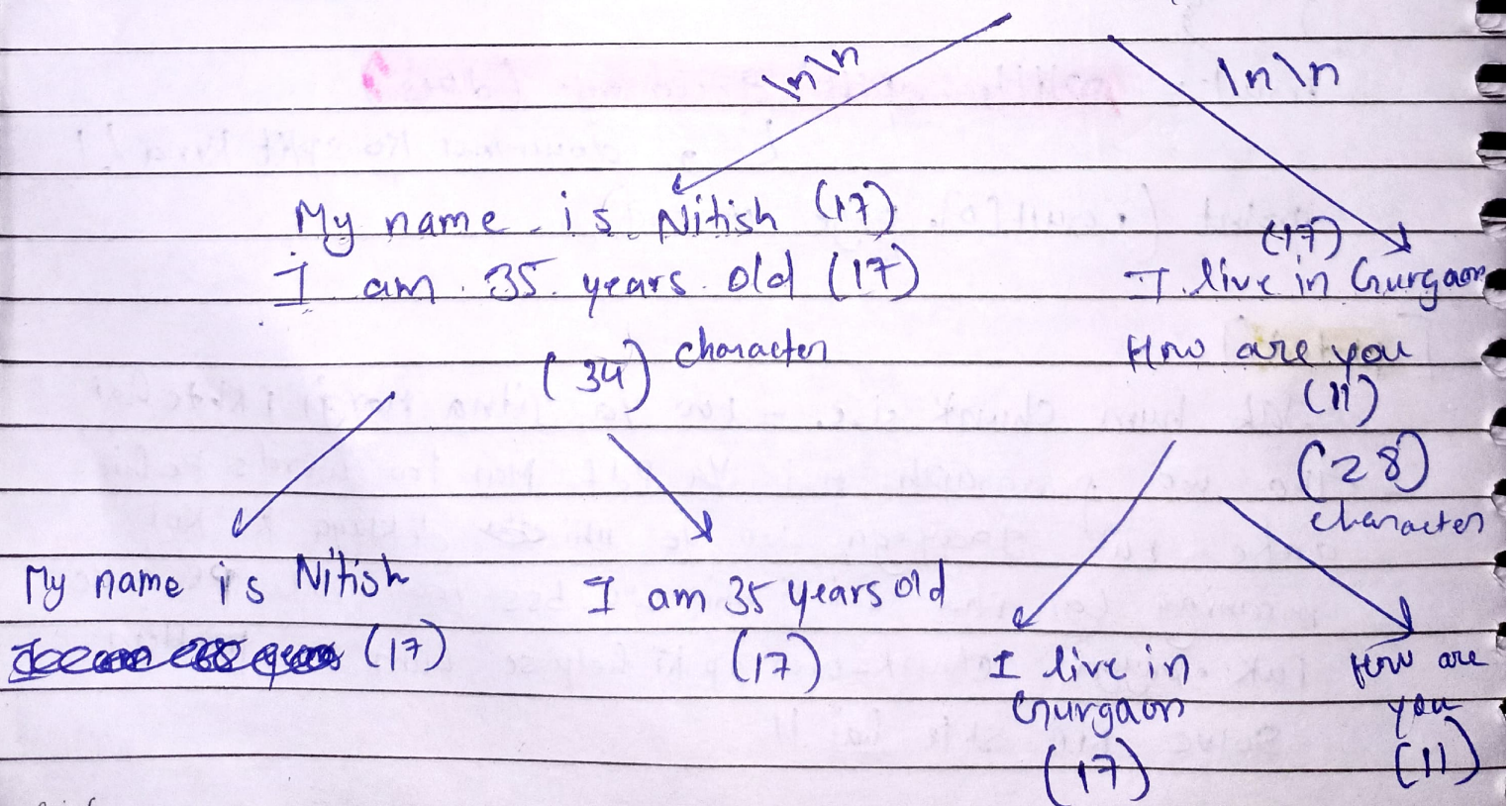
\n\n → paragraph
In → line (sentences)
' ' → words
" → character

for example

Allowed

[chunk-size = 10]

[My name is Nitish (17)
I am 35 years old (17)
I live in Gurgaon (17)
How are you (11)]



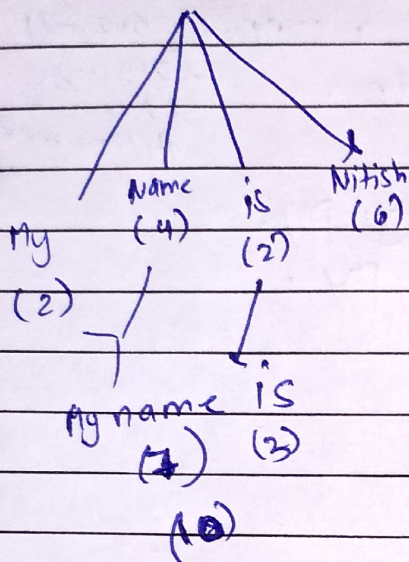
$$\begin{array}{r} 34 \\ 28 \\ \hline 62 \end{array}$$

$$\begin{array}{r} 54 \\ 28 \\ \hline 82 \end{array}$$

Date

My Name is Nitish

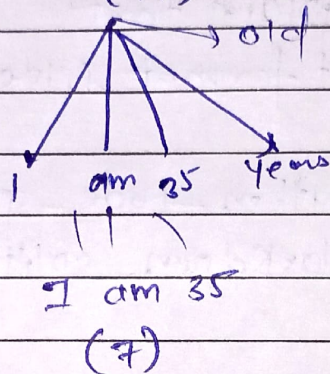
(17)



16 not more than 10 not include nitish.

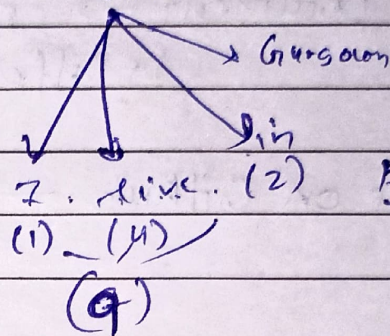
I am 35 years old

(17)



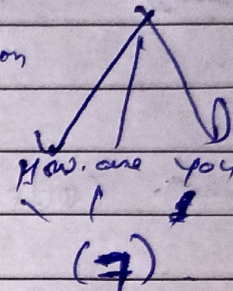
I live in Gurgaon

(17)



How are you

(11)



So,

My Name is Nitish
I am 35 years old
I live in Gurgaon
How are you

chunk-size = 10

[my name is] Nitish

[I am 35] years old

[I live in] Gurgaon

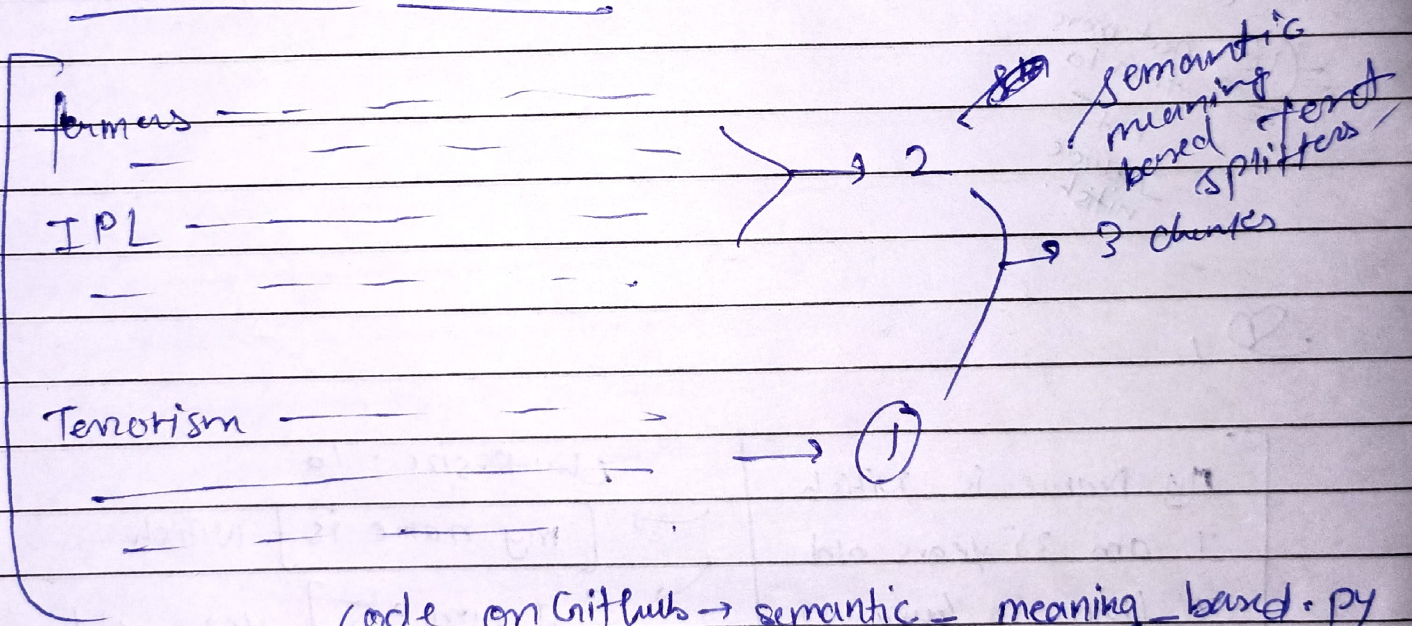
[How are] you

③ Document - Structure Based :

→ Document → no plain text → support overall language like python more.
↳ Different format like code.

→ Code on Github → python-code-splitting.py,
Markdown-splitting.py.

④ Semantic Meaning Based :



code on Github → semantic-meaning-based.py
use OPEN_API_KEY → ✓